

Diet, Nutrition & Sports Supplementation Platform — Complete Updated Product Documentation

Project Vision

Build a premium AI-driven Diet, Nutrition, Fitness, and Sports Supplementation ecosystem focused on:

- Personalized health transformation
- Monthly adaptive diet planning
- Smart nutrition recommendations
- Real-time progress tracking
- Family wellness management
- Gamification and retention
- Community engagement
- Expert consultations

The platform should emotionally motivate users while simplifying diet adherence and long-term consistency.

The experience should feel like:

- MyFitnessPal
- HealthifyMe
- Fitbit
- Cult Fit

But:

- More personalized
- More premium
- More community-driven
- More gamified
- More retention-focused

Technology Stack

Mobile Application

- Flutter
- GetX

- get_storage
- get_cli

Backend

- Node.js
- Express.js
- Sequelize ORM
- MySQL

Admin Panel

- Flutter Responsive Admin Panel

Third Party Services

- OneSignal Push Notifications
- Agora Video SDK
- Razorpay
- Stripe
- AWS S3 / Cloudinary
- Firebase Analytics
- Firebase Crashlytics

Core Product Flow

1. User installs app
 2. Registers/login
 3. Completes health onboarding
 4. Generates screening report
 5. Purchases personalized plan
 6. Starts plan whenever convenient
 7. Plan becomes active for 30 days from activation
 8. User follows meals and reminders
 9. Tracks progress daily
 10. Receives rewards and gamification points
 11. Re-checks weight after 30 days
 12. New adaptive plan generated according to progress
 13. Automatic expert consultation triggered if no progress observed
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Important Business Logic

Personalized Plan Activation System

Flow

- Diet plans are NOT auto-started immediately after purchase.
- User can start the generated plan anytime according to their convenience.
- Once the user taps: "Start My Plan" the 30-day duration begins.

Plan Validity

- Every generated plan remains active for 30 days only.
- Timer should be visible on dashboard.
- Countdown indicator:
 - Days left
 - Completion percentage

Plan Renewal

- After 30 days:
 - User must check updated weight
 - User must update progress data
 - New plan gets generated dynamically according to progress
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Pricing Model (Updated)

Remove Traditional Membership Plans

Do NOT use:

- Monthly subscription
- Quarterly subscription
- Yearly subscription

Instead use:

Per Plan Pricing Model

Example

- One personalized 30-day plan = ₹X

Benefits

- Easier onboarding
 - More transparent pricing
 - Better family expansion model
 - Better retention logic
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Family Module Business Logic

Family Add-On System

Users can:

- Add wife
- Husband
- Parents
- Children
- Siblings

Each member gets:

- Separate assessment
- Separate diet plan
- Separate reminders
- Separate progress tracking

Discount Logic

Example:

- Primary member plan = ₹X
- Additional family member = discounted price

Benefits:

- Encourages multi-user onboarding
 - Improves retention
 - Increases lifetime value
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Monthly Progress Evaluation System

Mandatory Progress Check

At the end of every 30-day cycle:

- User MUST update:
- Current weight
- Waist inches
- Progress photos (optional)

System compares:

- Old vs new weight
 - Inches lost/gained
 - BMI changes
 - Fat loss estimate
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Achievement Celebration System

Show emotional motivational messages.

Examples:

Weight Loss

"Hurray! You lost 2 kg of body fat!"

Inch Loss

"Amazing! You reduced 2 inches from your waist."

Consistency

"You completed 92% of your meal schedule this month!"

Streaks

"You're on a 21-day nutrition streak!"

Use:

- Confetti animations
 - Reward coins
 - Achievement badges
 - Progress graphs
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Smart Retention & Consultation Logic

No Progress Detection

If after 30 days:

- Weight remains unchanged OR
- Progress extremely low

Then:

Automatically:

- Book consultation slot suggestion
- Notify user
- Escalate to expert dashboard

User Notification Example

"We value your money and the hard work you put in to earn it. We don't want your efforts to go waste. We have scheduled a free consultation with our health expert so we can guide you personally and improve your results."

Consultation Features

- One-to-one expert guidance
- Free review session
- Personalized corrections
- Plan optimization

This becomes a major trust-building feature.

Adaptive Diet Engine

Monthly Dynamic Plan Generation

Every month:

- Plans should change dynamically.
- Meal structure should adapt according to:
- User progress
- Weight changes
- Goal completion
- Compliance score
- Food preferences

Example

Month 1:

- Basic calorie deficit

Month 2:

- Increased protein
- Different meal combinations
- Modified calories

Month 3:

- Advanced optimization
- Plateau handling

This avoids:

- Diet boredom
- User drop-off
- Repetitive meals

Social Room Module

Community Engagement System

Create a dedicated section:

Social Room

Users can:

- Share success stories
 - Upload transformation photos
 - Post achievements
 - Share diet journey
 - Motivate others
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Social Room Features

Post Types

- Transformation story
- Weight loss achievement
- Workout update
- Meal prep photo
- Motivation post

Interactions

Users can:

- Like posts
- Comment
- Share posts
- Follow users

Social Sharing

User can share their achievement externally:

- WhatsApp
 - Instagram
 - Facebook
 - X (Twitter)
 - Email
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Viral Growth Strategy

Every shared achievement should contain:

- User transformation summary
- App branding
- Download CTA

Example: "I lost 5 kg using FitFuel AI Nutrition App!"

This creates organic growth.

Main Mobile Application Modules

Splash Screen

Authentication

User Onboarding

Health Assessment

Screening Report

Smart Meal Planner

Supplement Recommendation

Smart Reminder System

Progress Tracking

Gamification

Wallet & Rewards

Family Management

Consultation Booking

Video Calling

Notifications

Health Tips

Social Room

Reports & Sharing

Profile & Settings

Health Assessment Engine

Generate:

- BMI
- BMR
- IBW
- %IBW
- TDEE

Classify:

- Underweight
- Normal
- Overweight
- Obese

Generate:

- Color coded report
 - Suggestions
 - Graphs
 - Goal recommendation
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Smart Meal Planner

Generate:

- Breakfast
- Mid meal
- Lunch
- Evening snack
- Dinner
- Water intake

Every item contains:

- Food name
- Calories
- Protein
- Carbs
- Fat
- Fiber
- Quantity
- Timing

Features:

- Replace food
 - Skip food
 - Edit quantity
 - Auto nutrition recalculation
 - Save favorites
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Supplement Recommendation Engine

Suggest according to:

- Goal
- Diet type
- Workout intensity
- Deficiency

Examples:

- Whey protein
 - Creatine
 - Fish oil
 - Multivitamin
 - Electrolytes
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Smart Reminder Engine

Examples: "Hey Rakesh, consume 2 egg whites now."

Reminder types:

- Meal reminders
- Water reminders
- Supplement reminders
- Weight check reminders
- Progress upload reminders

Features:

- Mark completed
 - Snooze
 - Auto repeat
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Gamification System

Features:

- Daily streaks
- XP points
- Reward wallet
- Achievement badges
- Meal completion score
- Leaderboards

Wallet redemption:

- Plan discounts
 - Consultation discounts
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Progress Tracking

Track:

- Weight
- Inches
- BMI history
- Compliance
- Water intake
- Meal adherence
- Streak history

Charts:

- Weekly
 - Monthly
 - Yearly
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Consultation System

Users can:

- Book appointments
- Join video consultation
- Pay consultation fees
- Receive expert advice

Experts can:

- Accept/reject slots
- Add notes
- Modify plan recommendations

Use:

- Agora SDK
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Social Room Screens

Feed Screen

- Trending stories
- Recent achievements
- Transformation highlights

Create Post Screen

- Upload images
- Add caption
- Share achievement

User Profile Social Tab

- User posts
 - Likes
 - Followers
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Admin Panel Modules

Dashboard Analytics

User Management

Family Management

Meal Plan Management

Formula Configuration

Supplement Management

Reminder Engine

Consultation Management

Reward Configuration

Social Room Moderation

Push Notification Management

Reports & Analytics

Admin Analytics

Track:

- Active users
 - Active plans
 - Renewal percentage
 - User retention
 - Weight loss achievements
 - Most successful plans
 - Revenue analytics
 - Family plan conversions
 - Social engagement analytics
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Backend Requirements

- REST APIs
 - JWT Authentication
 - Refresh Tokens
 - RBAC
 - Swagger APIs
 - Cron jobs
 - Notification queue
 - Reminder scheduler
 - Formula engine
 - Dynamic recommendation engine
 - Social room APIs
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Database Modules

- users
 - family_members
 - meal_plans
 - active_plans
 - plan_history
 - food_items
 - supplements
 - rewards
 - wallets
 - reminders
 - consultations
 - appointments
 - social_posts
 - social_likes
 - social_comments
 - notifications
 - progress_logs
 - transactions
 - formula_configurations
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UX Design Philosophy

The application should:

- Emotionally motivate users
- Build long-term consistency

- Reward discipline
- Reduce diet frustration
- Increase retention
- Encourage community participation

The UX should feel:

- Premium
 - Motivational
 - Scientific
 - Human
 - Easy to use
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Recommended Future Features

AI Nutrition Coach

Barcode Scanner

Wearable Integration

AI Food Image Recognition

Marketplace

Smart Grocery Suggestions

Recipe Generator

AI Chat Assistant

Final Product Goal

Create a scalable fitness-tech SaaS platform where:

- Users stay consistent
- Users see measurable transformation
- Users trust the platform
- Users invite family members
- Users share success stories publicly

- Experts can guide users effectively
- Retention naturally increases through personalization and community

The core strength of the platform should become:

- Personalized adaptive nutrition
- Emotional engagement
- Community motivation
- Progress visibility
- Smart retention automation